Western Governor's University

D213 Task 1 : Time Series Modeling

Brandon Velez

Student ID:012279120

10/9/2025

As a data analyst, you will assess continuing data sources for their relevance to specific research questions throughout your career.

In your previous coursework, you performed data cleaning and exploratory data analysis on your data. You have seen basic trends and will now build more sophisticated statistical models.

For this task, you will select one of the data sets and associated data dictionaries from "Data Files and Associated Dictionary Files" in the Web Links section.

Part I: Research Question

A. Describe the purpose of this data analysis by doing the following:

1. Summarize one research question that is relevant to a real-world organizational situation captured in the selected data set and that you will answer using time series modeling techniques.

Can the given dataset showing revenue over time be implemented in a prediction model to provide a forecast of revenue for the next quarter?

2. Define the objectives or goals of the data analysis. Ensure your objectives or goals are reasonable within the scope of the scenario and are represented in the available data.

This data analysis aims to use an ARIMA model to output a forecast of revenue for the ninety-day quarter directly after the dataset ends.

Part II: Method Justification

B. Summarize the assumptions of a time series model including stationarity and autocorrelated data.

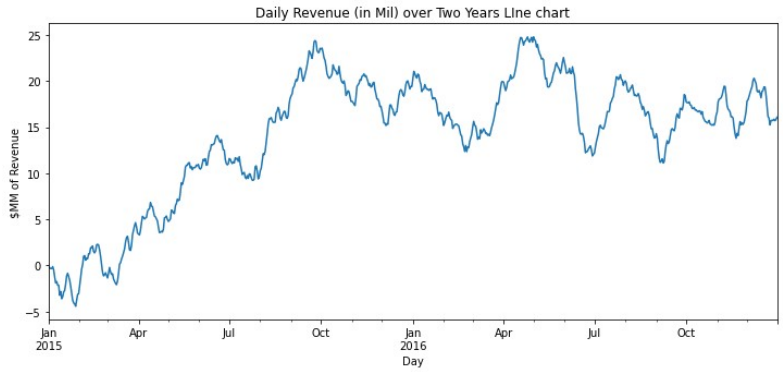
A time series model assumes that data is stationary, meaning long term trends over time have been removed (visually flattened). This model also assumes that autocorrelation remains constant over the relevant time period.

(“6.4.4.2. Stationarity”)

Part III: Data Preparation

C. Summarize the data cleaning process by doing the following:

1. Provide a line graph visualizing the realization of the time series.

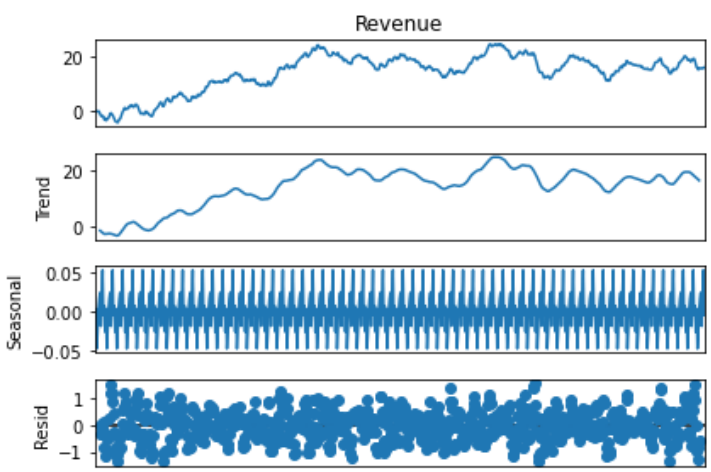


2. Describe the time step formatting of the realization, including any gaps in measurement and the length of the sequence.

The data dictionary states that this time series consists of 731 total days, with daily data recordings. Day one records a revenue of zero, but there are no other gaps apparent in the data.

3. Evaluate the stationarity of the time series.

Prior to decomposing the dataset, this data is not stationary. However, when decomposition is implemented, stationarity is achieved, as illustrated by the following visualizations of the transformed dataset. The 'seasonal' graph now shows a consistent range for the data values and residuals show no clear pattern.



4. Explain the steps you used to prepare the data for analysis, including the training and test set split.

First, the dataset has a .dropna function run to make sure any not available values will not impede calculations.

Then, the dataset is datetime indexed using pandas date_range function to allow calculations to properly show change over time. An arbitrary start date of 2015-1-01 is selected.

A train set dataframe is created containing the first 701 daily measurements, and a test set dataframe defined as the 30 most recent measurements.

5. Provide a copy of the cleaned data set.

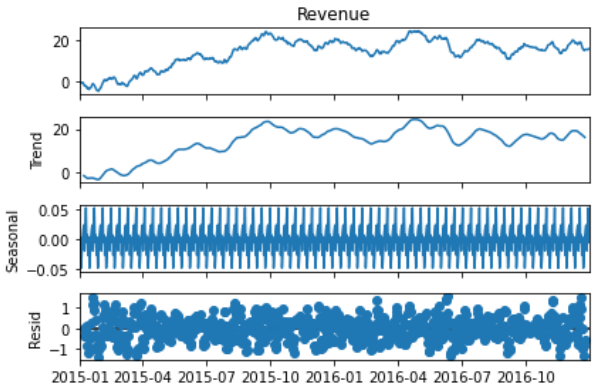
See file attached: “CleanData.csv”

Part IV: Model Identification and Analysis

D. Analyze the time series data set by doing the following:

1. Report the annotated findings with visualizations of your data analysis, including the following elements:

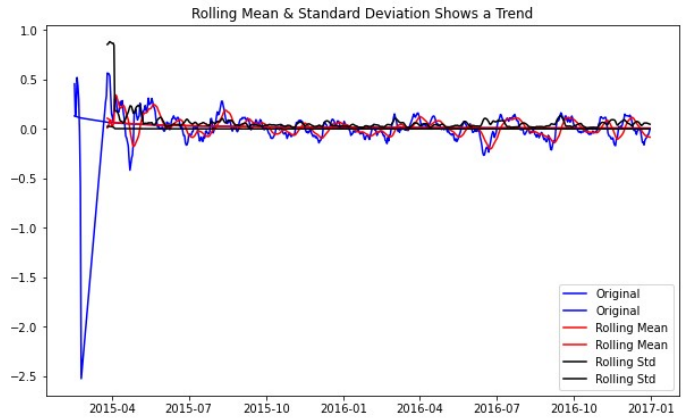
- the presence or lack of a seasonal component



The cyclical nature of the seasonal plot confirms that the revenue dataset is seasonal.

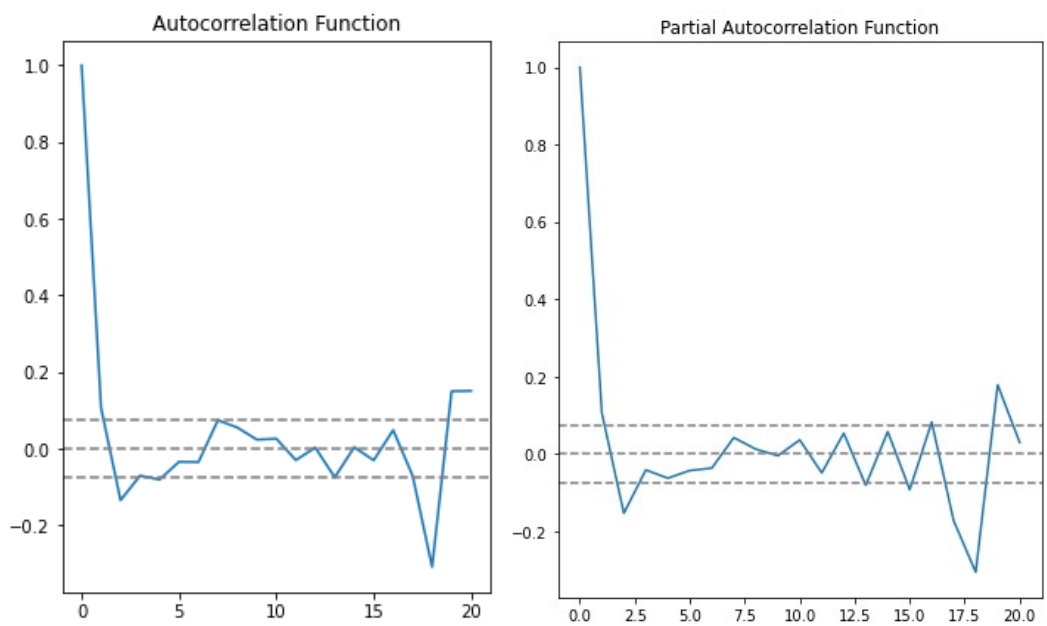
- trends

The trends plot above indicates that there are trends relevant in the dataset.



The above chart visualizing rolling mean and standard deviation versus the original dataset also seems to visually indicate clear cyclical highs and lows within the data. During the first few months, likely the hospital's ramp up period, the data is much more difficult to find these trends in, but once things stabilize after a few months these trends appear very consistent.

- the autocorrelation function

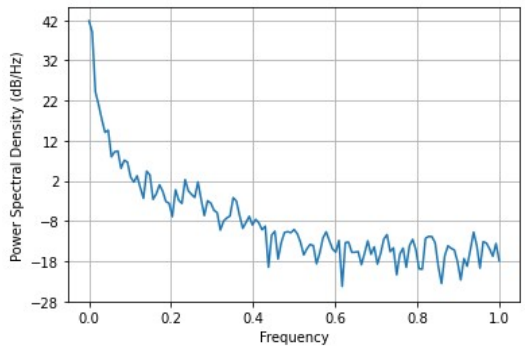


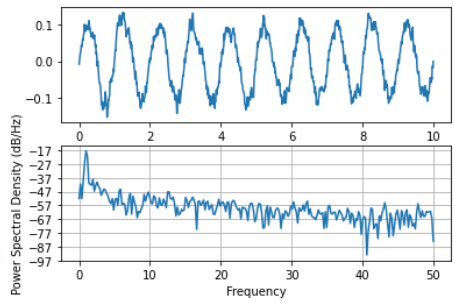
Both the autocorrelation function and partial autocorrelation function plots show that after the ramp up period, revenue values grow to be relatively consistent and predictable. Autocorrelation is a measure of how well a current value correlates with the prior values in the time series. Partial autocorrelation is calculated by finding the correlation within the transformed time series as before.

Stable correlation within the range of -1 to 1 is seen for most of the observed time period.

(“10.2 - Autocorrelation and Time Series Methods | STAT 462”)

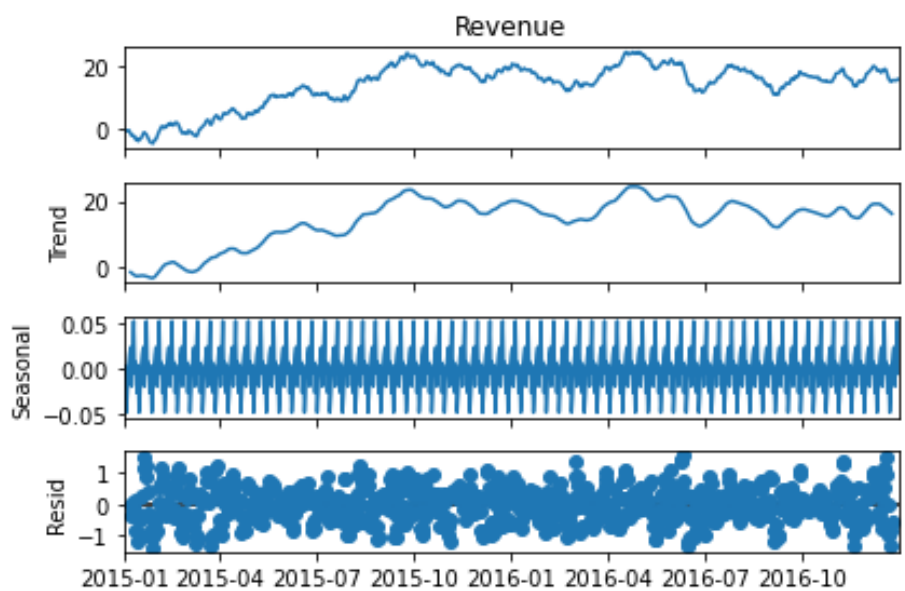
- the spectral density





The spectral density graphs show a clear seasonal trend as there are many recurring cycles.

- the decomposed time series



This decomposed time series graph showing revenue over time illustrates that there are seasonal cycles present.

- confirmation of the lack of trends in the residuals of the decomposed series

There is no clear pattern among the residuals of the ARIMA analysis, adding validity to the results.

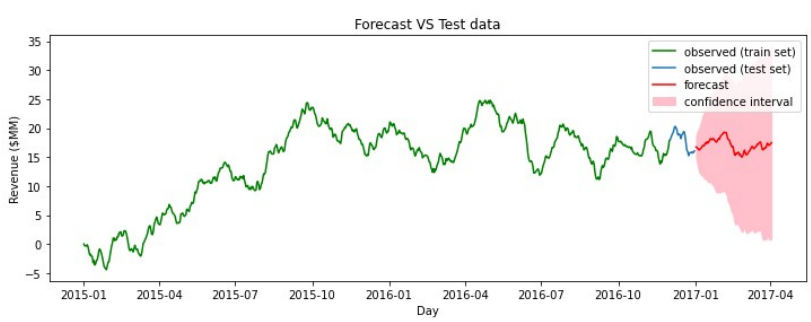
2. Identify an autoregressive integrated moving average (ARIMA) model that accounts for the observed trend and seasonality of the time series data.

```
   ...:
   ...: from pmdarima import auto_arima
   ...: import warnings
   ...: warnings.filterwarnings("ignore")
   ...: stepwise_fit=auto_arima(df['Revenue'], trace=True, suppress_warnings=True)
   ...: stepwise_fit.summary()
Performing stepwise search to minimize aic
 ARIMA(2,1,2)(0,0,0)[0] intercept   : AIC=883.277, Time=0.75 sec
 ARIMA(0,1,0)(0,0,0)[0] intercept   : AIC=1015.972, Time=0.09 sec
 ARIMA(1,1,0)(0,0,0)[0] intercept   : AIC=881.359, Time=0.11 sec
 ARIMA(0,1,1)(0,0,0)[0] intercept   : AIC=906.199, Time=0.11 sec
 ARIMA(0,1,0)(0,0,0)[0]             : AIC=1015.481, Time=0.05 sec
 ARIMA(2,1,0)(0,0,0)[0] intercept   : AIC=883.300, Time=0.20 sec
 ARIMA(1,1,1)(0,0,0)[0] intercept   : AIC=883.314, Time=0.19 sec
 ARIMA(2,1,1)(0,0,0)[0] intercept   : AIC=883.348, Time=0.39 sec
 ARIMA(1,1,0)(0,0,0)[0]             : AIC=879.982, Time=0.06 sec
 ARIMA(2,1,0)(0,0,0)[0]             : AIC=881.911, Time=0.07 sec
 ARIMA(1,1,1)(0,0,0)[0]             : AIC=881.927, Time=0.08 sec
 ARIMA(0,1,1)(0,0,0)[0]             : AIC=905.166, Time=0.05 sec
 ARIMA(2,1,1)(0,0,0)[0]             : AIC=881.947, Time=0.19 sec

Best model:  ARIMA(1,1,0)(0,0,0)[0]
Total fit time: 2.376 seconds
```

```
Out[6]:
<class 'statsmodels.iolib.summary.Summary'>
"""
                               SARIMAX Results
==============================================================================
Dep. Variable:                      y   No. Observations:                  731
Model:               SARIMAX(1, 1, 0)   Log Likelihood                -437.991
Date:                Wed, 15 Oct 2025   AIC                            879.982
Time:                        21:11:32   BIC                            889.168
Sample:                    01-01-2015   HQIC                           883.526
                         - 12-31-2016
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ar.L1          0.4142      0.034     12.258      0.000       0.348       0.480
sigma2         0.1943      0.011     17.842      0.000       0.173       0.216
===================================================================================
Ljung-Box (L1) (Q):                   0.02   Jarque-Bera (JB):                 1.92
Prob(Q):                              0.90   Prob(JB):                         0.38
Heteroskedasticity (H):               1.00   Skew:                            -0.02
Prob(H) (two-sided):                  0.97   Kurtosis:                         2.75
===================================================================================
```

3. Perform a forecast using the derived ARIMA model identified in part D2.



4. Provide the output and calculations of the analysis you performed.

5. Provide the code used to support the implementation of the time series model.

See file attached: “D213BrandonVelezTask1.pdf”

Part V: Data Summary and Implications

E. Summarize your findings and assumptions by doing the following:

1. Discuss the results of your data analysis, including the following points:

- the selection of an ARIMA model

The implemented ARIMA model was built by selecting the recommended P, D and Q values of 1,1,0 from running the auto ARIMA model over original dataset.

- the prediction interval of the forecast

Ninety days has been selected for the forecast period.

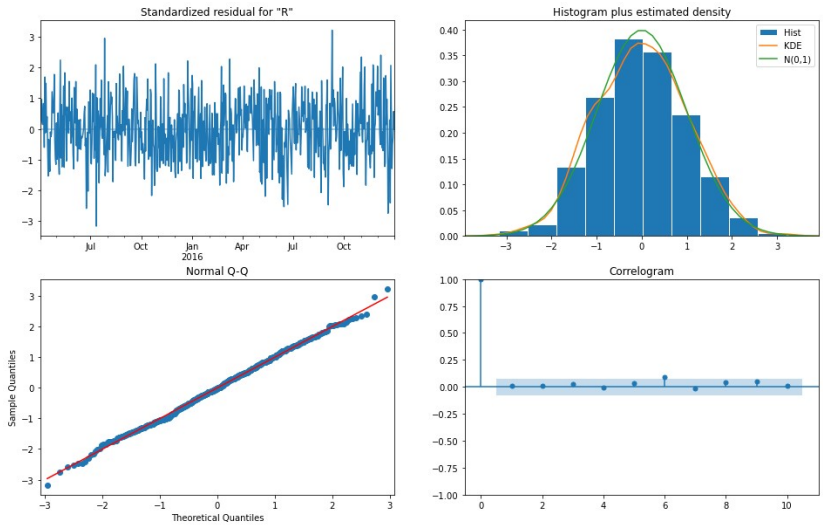
- a justification of the forecast length

Ninety days should give enough time to allow the company time to interview and go through the hiring process if new employees are necessary, ample time to prepare any supply orders, and give a strong estimate of the company’s revenue to inform management if they will be able to expand facilities and equipment and begin those processes. Going beyond one quarter would begin to diminish the accuracy of the predictions while looking beyond the scope of current business operations.

- the model evaluation procedure and error metric

```
In [91]:
    ...:
    ...: mae = np.mean(np.abs(results.resid))
    ...: print("MAE", mae)
MAE 0.452307939667293
```

First, the final ARIMA model was evaluated by calculating a mean absolute error score. This model scores 0.4523, indicating the average distance between the observed and predicted values is very close and indicates the model is making predictions accurately.



The final ARIMA model is also evaluated using a results.plot_diagnostics function, populating the above graphs. The standardized residual for R graph stays within a tight consistent range. The sample vs theoretical quantities line graph has the samples and theoretical values lined up very closely. These are more indicators that the model is predicting values closely to the observed values.

```
In [13]:
    ...:
    ...: from sklearn.metrics import mean_squared_error
    ...: from math import sqrt
    ...: rmse = sqrt(mean_squared_error(pred,test['Revenue']))
    ...: print(rmse)
1.7676801786810543
```

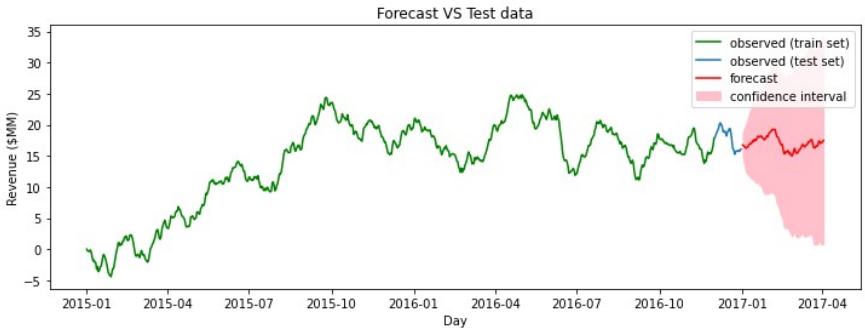
The revenue predictions generated using the test dataset can be compared to the actual revenue values in the test set to provide an RMSE of 1.76. A low RMSE score indicates that the mean difference between prediction values and actual values is low and indicates a strong model. This is another positive indicator for the implemented ARIMA models.

(Zach Bobbitt)

2. Provide an annotated visualization of the forecast of the final model compared to the test set.

The following graph first shows the two years of observed revenue, broken into test as the first 701 days in green, and train as the final 30 observations in

blue. The graph then shows predicted revenue in red based on the forecast predictions provided by the ARIMA model built using the recommended P, D, and Q values. A confidence interval is also provided by running upper and lower confidence interval functions. An arbitrary start date of 2015-1-01 was selected for the observed data, so the observations end on the final day of 2016 to represent two years. The first three months of 2017, directly following the observations, are used for the forecast time period. Visually, the forecast data appears to remain stable and consistent with the ranges of the previous year and a half. The earliest six to seven months of the observed revenue dataset appear to be a ramp up period so this is less relevant than the more recent data for forecasting.



3. Recommend a course of action based on your results.

Based on the ARIMA model forecast, it would be recommended that hospital management use the provided revenue forecast for the next quarter to plan hiring, supplying, and growth for project planning during this time period.

Part VI: Reporting

F. With the information from part E, create your report using an industry-relevant interactive development environment (e.g. a Jupyter Notebook). Include a PDF or HTML document of your executed notebook presentation.

G&H: Sources

Works Cited

"6.4.4.2. Stationarity." *Nist.gov*, 2019, www.itl.nist.gov/div898/handbook/pmc/section4/pmc442.htm.

"10.2 - Autocorrelation and Time Series Methods | STAT 462." *Online.stat.psu.edu*, online.stat.psu.edu/stat462/node/188/.

"Pandas.DataFrame.drop — Pandas 1.3.2 Documentation." *Pandas.pydata.org*, pandas.pydata.org/pandas-docs/stable/reference/api/pandas.DataFrame.drop.html.

"Pandas.DataFrame.plot — Pandas 1.1.2 Documentation." *Pandas.pydata.org*, pandas.pydata.org/pandas-docs/stable/reference/api/pandas.DataFrame.plot.html.

Zach Bobbitt. "How to Interpret Root Mean Square Error (RMSE)." *Statology*, 10 May 2021, www.statology.org/how-to-interpret-rmse/.